Zachary Kenneth Fisher

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Staff Software Engineer at Google DeepMind building frontier LLM post-training systems, reward models, synthetic data pipelines, and scientific-agent infrastructure. Ph.D. physicist with a track record of rigorous research, production ML launches, and simulator-backed evaluation of model capabilities.

Employment

Google / Google DeepMind — *Mountain View, CA*
Staff Software Engineer — *May 2025 – Present*
Senior Software Engineer — *April 2022 – April 2025*
Software Engineer — *April 2020 – April 2022*

- Built reward and value-modeling systems for reinforcement learning post-training of frontier LLMs, including rubric-conditioned critiques, deterministic code-based auto-raters, and feedback-driven critic improvement methods.
- Designed and scaled a failure-mode-driven synthetic data generation pipeline for frontier model post-training across SQL, data science, mathematical reasoning, and scientific coding. The system generated targeted training curricula with automated multi-stage quality filtering and simulator-backed verification.
- Drove adoption of synthetic training data into multiple Gemini training revisions, contributing to multi-point improvements on mathematical and tool-use benchmarks.
- Diagnosed and resolved critical failures in frontier training runs, including silent infrastructure failures, reward signal degradation, evaluation flakiness, and distribution shifts in model behavior. Built real-time health dashboards and sample-exploration tools to accelerate live-run triage.
- Conceived and led development of a scientific-agent benchmark for long-horizon optimization with real-world simulators. Designed dozens of tasks across engineering and science domains, built the evaluation infrastructure and leaderboard from scratch, evaluated 7+ frontier models, and identified a 2–4× exploration gap between leading model families.
- Co-authored research on simulator-driven reinforcement learning for agentic tool use, and designed a procedural data generation framework producing dense-reward inverse problems across 5 scientific domains for RL post-training.
- Led work on attribute extraction models for Google Shopping and Maps, processing tens of millions of edits across 80+ attributes at over 96% precision and improving multilingual coverage by 3–5× recall across four languages.
- Contributed to LLM-powered systems in AI Overviews and Shopping through tool-use evaluation, reward modeling, and structured extraction supporting user-facing launches.
- Enabled partner teams across Search, Maps, and post-training by sharing models, evaluation frameworks, and infrastructure. Mentored engineers and researchers across organizations and taught an internal machine learning course.

Perimeter Institute for Theoretical Physics — *Waterloo, ON, Canada*
Postdoctoral Researcher — *September 2017 – February 2020*

- Conducted research in quantum field theory, quantum information, and quantum gravity as an It from Qubit postdoctoral fellow. Published work on black hole information, holography, entropy bounds, and quantum energy conditions.
- Co-formulated the Quantum Null Energy Condition in work on the Quantum Focusing Conjecture and co-authored its first proof, connecting quantum information inequalities to spacetime energy conditions.

Education

University of California, Berkeley *August 2012 – May 2017*
Ph.D. in Physics. Thesis: "Entropy Bounds and Entanglement."

Massachusetts Institute of Technology *September 2008 – June 2012*
S.B. in Physics. Thesis: "Shuttling of ions for characterization of a microfabricated ion trap."

Selected Publications

- H. Wu, **Z. Fisher** et al. "Atomated Optimization of Structured Large Language Model Automated Optimization of Structured Large Language Model Prompts via Iterative Evaluation." Technical Disclosure Commons (2026).
- Y. Sheng, S. Gandhe, B. Kanagal, N. Edmonds, **Z. Fisher**, S. Tata, A. Selvan. "Measuring an LLM's Proficiency at Using APIs: A Query Generation Strategy." KDD-ADS (2024).
- R. Aksitov, S. Miryoosefi, Z. Li, D. Li, S. Babayan, K. Kopparapu, **Z. Fisher**, R. Guo, S. Prakash, P. Srinivasan, M. Zaheer, S. Kumar. "ReST meets ReAct: Self-Improvement for Multi-Step Reasoning LLM Agent." arXiv:2312.10003.
- B. Gunel, J.B. Wendt, J. Xie, Y. Zhou, N. Vo, **Z. Fisher**, S. Tata. "STRUM-LLM: Attributed and Structured Contrastive Summarization." arXiv:2403.19710.
- S. Ontanon, J. Ainslie, V. Cvicek, **Z. Fisher**. "Making Transformers Solve Compositional Tasks." ACL (2022).
- J. Ainslie, S. Ontanon, C. Alberti, V. Cvicek, **Z. Fisher**, P. Pham, A. Ravula, S. Sanghai, Q. Wang, L. Yang. "ETC: Encoding Long and Structured Inputs in Transformers." EMNLP (2020).
- R. Bousso, **Z. Fisher**, S. Leichenauer, A. Wall. "Quantum Focusing Conjecture." Phys. Rev. D 93, 064044 (2016).
- R. Bousso, **Z. Fisher**, J. Koeller, S. Leichenauer, A. Wall. "Proof of the Quantum Null Energy Condition." Phys. Rev. D 93, 024017 (2016).
- H. Z. Chen, **Z. Fisher**, J. Hernandez, R. C. Myers, S. M. Ruan. "Information Flow in Black Hole Evaporation." Journal of High Energy Physics 3 (2020).

Awards

- Product Achievement Award and Publication Excellence Award, Google Research / Google DeepMind.
- It from Qubit Postdoctoral Fellowship, Simons Collaboration on Quantum Fields, Gravity and Information.
- Editors' Suggestion, Physical Review D, for work on quantum entropy bounds.

Skills

- **Core engineering:** Python, JAX, PyTorch, TensorFlow, Apache Beam, SQL, C++, distributed training, production ML debugging.
- **ML systems:** RL post-training, RLHF, reward modeling, value modeling, model evaluation, agent evaluation, tool use, synthetic data generation, long-context agents.
- **Research domains:** scientific AI, simulator-backed evaluation, quantum information, quantum gravity, entropy bounds, mathematical reasoning.